

SETS — ADVANCED CHALLENGE PAPER

ICSE Mathematics · Class VII · Chapter: Sets — Survey Problems, Venn Diagrams & Identities

Total Marks: 67

Time Allowed: 1¾ Hours

General Instructions:

- All questions are compulsory. Marks for each question are shown in brackets.
- For three-set survey problems, work from the innermost region (all three) outward, and show your working.
- For Venn diagram questions, shade neatly using a pencil.

Section A — Verifying Set Identities

(8 marks)

A1. If $U = \{1, 2, 3, \dots, 12\}$ is the universal set, $A = \{1, 4, 6, 9, 10\}$, and $B = \{2, 4, 7, 9, 11\}$, verify that $(A \cup B)' = A' \cap B'$. [4]

A2. If $A = \{2, 4, 6, 8, 10, 12, 14, 16\}$, $B = \{4, 6, 10, 14\}$, and $C = \{2, 6, 12, 16\}$, verify that $A - (B \cup C) = (A - B) \cap (A - C)$. [4]

Section B — Two-Set Survey Problems

(13 marks)

B1. In a class of 45 students, every student likes at least one of two subjects — Mathematics or Science. If 28 students like Mathematics and 25 like Science, how many students like **only** Mathematics? [3]

B2. A survey of 500 people found that 380 like cricket and 290 like football. What is the **least** number of people who must like both sports? [3]

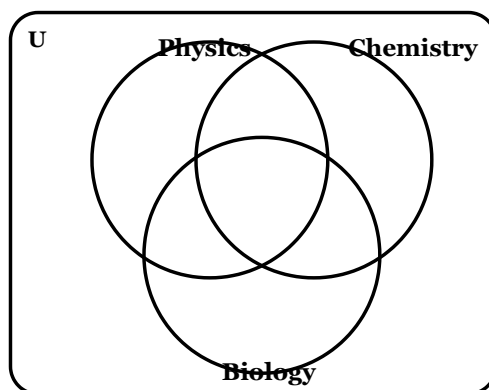
B3. Let A and B be two finite sets such that $n(A - B) = 18$, $n(A \cup B) = 95$, and $n(A \cap B) = 32$. Find $n(A)$ and $n(B)$. [4]

B4. If X and Y are two sets such that $n(X) = 24$, $n(Y) = 31$, and $n(X \cup Y) = 48$, find $n(X \cap Y)$. [3]

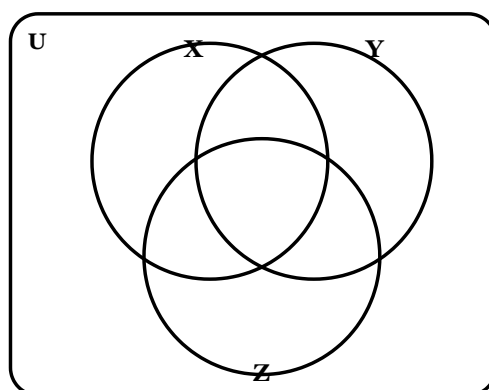
Section C — Three-Set Survey Problems**(16 marks)**

*Tip: start with the number who study/read/like **all three**, then work outward to find each "only" region, using the blank diagram provided.*

- C1.** In a survey of 100 students, 45 study Physics, 38 study Chemistry, and 40 study Biology. 15 study both Physics and Chemistry, 12 study both Chemistry and Biology, 10 study both Physics and Biology, and 5 study all three subjects. Find: (i) how many study Biology only (ii) how many study Physics and Biology but not Chemistry (iii) how many study at least one of the three subjects. [8]



- C2.** In a survey of 80 people, 30 read magazine X, 26 read magazine Y, and 28 read magazine Z. 10 read both X and Y, 8 read both Y and Z, 12 read both X and Z, and 4 read all three magazines. Find: (i) the number of people who read at least one magazine (ii) the number of people who read exactly one magazine. [8]

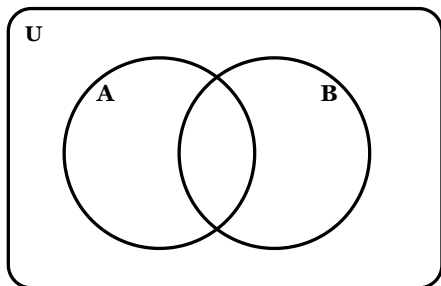
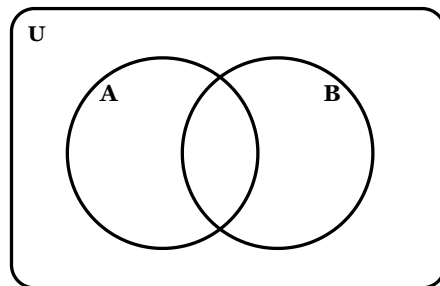
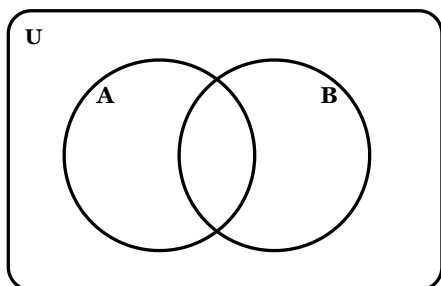
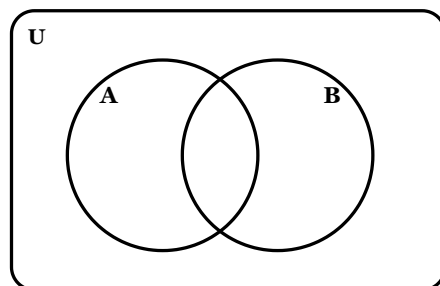
**Section D — Cartesian Product****(3 marks)**

*Recall: the Cartesian product $A \times B$ is the set of **all ordered pairs** (a, b) where $a \in A$ and $b \in B$.*

- D1.** If $A \times B = \{(x,1), (x,2), (y,1), (y,2)\}$, find the sets A and B. [3]

Section E — Venn Diagrams: A Visual Proof of De Morgan's Laws (12 marks)

Shade each diagram below for the given expression. Then compare $E1$ with $E2$, and $E3$ with $E4$ — what do you notice about each pair?

E1. Shade $(A \cup B)'$ **E2. Shade $A' \cap B'$** **E3. Shade $(A \cap B)'$** **E4. Shade $A' \cup B'$** 

Section F — Quick Set Theory Questions**(16 marks)**

F1. What type of set is $\{x : x \in \mathbb{N} \text{ and } x + 5 = 3\}$? Justify your answer. [2]

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F2. Write the set $A = \{x : x \in \mathbb{Z}, -3 < x \leq 4\}$ in roster form. [2]

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F3. From the sets given below, select **all** pairs of equal sets:

$A = \{3, 6, 9, 12\}$, $B = \{5, 6, 7, 8\}$, $C = \{9, 12, 3, 6\}$, $D = \{8, 5, 6, 7\}$, $E = \{-3, 3\}$, $F = \{0, k\}$, $G = \{3, -3\}$, $H = \{0, 1\}$. [3]

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F4. If $U = \mathbb{N}$ (the set of natural numbers) and $A = \{x : x \text{ is a multiple of } 5, x \in \mathbb{N}\}$, describe A' in words. [2]

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F5. Show, with a numerical example, that $A \cap B = A \cap C$ does **not** necessarily mean $B = C$. [3]

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F6. In a town of 600 people, 280 speak Tamil, 210 speak Telugu, and 90 speak both languages. Find the number of people who speak **neither** language. [4]

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